

# Deep Learning for MSc Crib Sheet

February 19, 2026

## 1 Formulae

$$a(z) = \frac{1}{1 + \exp(-z)}$$

$$\frac{da}{dz} = a(1 - a)$$

$$a(z) = \frac{\exp(z) - \exp(-z)}{\exp(z) + \exp(-z)}$$

$$\frac{da}{dz} = (1 - a^2)$$

$$a(z) = \max(0, z)$$

$$\frac{da}{dz} = \begin{cases} 0 & \text{if } z < 0 \\ 1 & \text{if } z \geq 0 \end{cases}$$

$$W_{out} = \frac{W_{in} - F + 2P}{S} + 1$$